

Building a College Student Portal

Code -

```
# ----- Decorator 1 -----
def login_required(func):
    def wrapper(*args, **kwargs):
        print("Login Successful")
        return func(*args, **kwargs)
    return wrapper

# ----- Decorator 2 -----
def activity_logger(func):
    def wrapper(*args, **kwargs):
        print(f"Activity Logged: Viewing profile of {args[0].name}")
        return func(*args, **kwargs)
    return wrapper

# ----- Student Class -----
class Student:

    def __init__(self, name, roll_no, student_id, branch):
        self.name = name
        self.roll_no = roll_no
        self.student_id = student_id
        self.branch = branch

    @login_required
    @activity_logger
    def show_profile(self):
        print("\n----- Student Profile -----")
        print("Name      :", self.name)
        print("Roll No   :", self.roll_no)
        print("Student ID :", self.student_id)
        print("Branch    :", self.branch)
```

```

print("-----")

# ----- Closure -----
def make_greeting(message):
    def greet(student):
        print(message + student.name)
    return greet

# ----- Creating Objects -----
student1 = Student("Rahul", 101, "MIT101", "Computer Engineering")
student2 = Student("Priya", 102, "MIT102", "Cybersecurity")

# ----- Display Profiles -----
student1.show_profile()
print()

student2.show_profile()
print()

# ----- Greeting Closures -----
welcome = make_greeting("Welcome ")
good_morning = make_greeting("Good Morning ")
good_evening = make_greeting("Good Evening ")

# ----- Using Closures -----
welcome(student1)
good_morning(student2)
good_evening(student1)

```

Output -

Output

```
^ Login Successful
Activity Logged: Viewing profile of Rahul

----- Student Profile -----
Name      : Rahul
Roll No   : 101
Student ID : MIT101
Branch    : Computer Engineering
-----

Login Successful
Activity Logged: Viewing profile of Priya

----- Student Profile -----
Name      : Priya
Roll No   : 102
Student ID : MIT102
Branch    : Cybersecurity
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Welcome Rahul
Good Morning Priya
Good Evening Rahul

=== Code Execution Successful ===
```